Bounds for generating pairs from maximal subgroups

Human Compression project: a component of the L046 audit

Abstract

A union bound over maximal subgroups gives a lower bound for the number of generating-pair orbits, without the full subgroup lattice. Applied to the O'Nan group with the published maximal-subgroup classification and ordinary character tables, it gives at least 798370 orbits in the convention of Connor and Leemans. This exceeds the total 796772 in their Theorem 1.1(2). One table entry can be determined exactly: the type $(20, 31)$ has 9118 orbits, rather than the printed 7246. The same calculation proves existence for all 129 types asserted to exist in that paper. We do not determine the exact total or the full chirality classification.

1 The counting convention and the bound

For a finite group G, let

$$X_G(p,q) = \{(R,S) : |R| = p,\ |S| = q,\ |RS| = 2\},$$
$$E_G(p,q) = \{(R,S) \in X_G(p,q) : \langle R, S\rangle = G\}.$$

Write their cardinalities as $x_G(p,q)$ and $e_G(p,q)$, and write $a = |\operatorname{Aut}(G)|$. Automorphisms act freely on $E_G(p,q)$: fixing both generators fixes every element of G. Consequently the number of orbits is $n_G(p,q) = e_G(p,q)/a$.

Pairs are ordered. Taking $p \le q$ selects one orientation of each type and makes no further duality identification when $p = q$. This is precisely the source's convention in its product-one triples and formula (2.4). Although the source also writes braces for pairs, the result below concerns the count defined by that formula and reported in Table 1 and Theorem 1.1(2).

Theorem 1. *Let $M_1, \dots, M_s$ represent all conjugacy classes of maximal proper subgroups of G. Put*

$$B_G(p,q) = \sum_{j=1}^{s} [G : M_j] x_{M_j}(p,q).$$

Then

$$\max\left(0, \left\lceil \frac{x_G(p,q) - B_G(p,q)}{a} \right\rceil\right) \ \le\ n_G(p,q) \ \le\ \left\lfloor \frac{x_G(p,q)}{a} \right\rfloor. \tag{1}$$

Proof. Every pair that does not generate G is contained in a maximal proper subgroup, because G is finite. A conjugate of M_j contains precisely $x_{M_j}(p,q)$ pairs of the prescribed type. The number of its conjugates is $[G : N_G(M_j)]$, which is at most $[G : M_j]$. Taking the union of these sets, and allowing repeated counting in intersections, gives

$$|X_G(p,q) \setminus E_G(p,q)| \le B_G(p,q).$$

Subtract from $x_G(p,q)$ and divide by a. The number of generating orbits is a nonnegative integer, giving the lower bound. The inclusion $E_G(p,q) \subseteq X_G(p,q)$ gives the upper bound, again using integrality. No assumption that maximal subgroups intersect trivially is made. Repeated isomorphism types in the list must retain their conjugacy-class multiplicities. □

The character calculation uses the Frobenius identity already acquired in the separately reviewed manuscript *Generating pairs from characters and subgroup incidence*. If C_i are the conjugacy classes of H, define $v_{\chi,t} = \sum_{|c_i|=t} |C_i|\chi(c_i)$. Then

$$x_H(p,q) = \frac{1}{|H|} \sum_{\chi \in \mathrm{Irr}(H)} \frac{v_{\chi,2} v_{\chi,p} v_{\chi,q}}{\chi(1)}. \tag{2}$$

The acquired proof explains the bijection with product-one triples. All arithmetic in (2) is exact. In the retained tables every sum $v_{\chi,t}$ is an integer, verified before export. A separate Python program then evaluates (2) with rational numbers from these integer vectors. Neither source generating-pair counts nor expected totals are inputs to (2) or (1).

2 The retained subgroup input

Let G be the O'Nan sporadic simple group. We retain its established order 460815505920, the identification $\mathrm{Aut}(G) = G : 2$, and the complete classification of its thirteen conjugacy classes of maximal subgroups. The following table records their isomorphism types, orders and multiplicities. The names in the last column identify the actual CTblLib tables used; distinct names can describe isomorphic groups.

type	order	classes	character tables
$L_3(7) : 2$	3753792	2	`L3(7).2`, `ONM2`
J_1	175560	1	`J1`
$4_2.L_3(4) : 2_1$	161280	1	`4_2.L3(4).2_1`
$(3^2 : 4 \times A_6).2$	25920	1	`ONM5`
$3^4 : 2^{1+4}.D_{10}$	25920	1	`3^4:2^(1+4)D10`
$L_2(31)$	14880	2	`L2(31)`, `ONM8`
$4^3.L_3(2)$	10752	1	`4^3.L3(2)`
M_{11}	7920	2	`M11`, `ONM11`
A_7	2520	2	`A7`, twice

The product notation for the order-25920 subgroup follows the online Atlas display; the exact named character table, not this abbreviation, is the finite input used in the computation.

These are external classification premises. We acquire the complete list from the Atlas of Group Representations, cross-checked against CTblLib's stored maximal-class list. Wilson's survey, Section 3.3, attributes independent determinations to Yoshiara, Ivanov–Tsaranov– Shpectorov and Wilson. We do not reproduce those classification proofs. The named character tables are likewise retained Atlas/CTblLib data; checking their orders or character identities alone would not prove their group identifications. The missing full subgroup-incidence matrix of G is not required for the weaker union bound (1).

3 The two numerical conclusions

Use exactly the source's set of rotation orders

$$P = \{3, 4, 5, 6, 7, 8, 10, 11, 12, 14, 15, 16, 19, 20, 28, 31\}.$$

For each of its 136 pairs $p \le q$, let $l(p,q)$ be the lower bound in (1). The complete ledger records x_G, each of the thirteen terms of B_G, and both integer bounds. The sums of $l(p,q)$ with a fixed first coordinate are as follows.

p	row lower sum	p	row lower sum
3	274	12	29404
4	5124	14	36418
5	7712	15	43644
6	18456	16	228833
7	29128	19	88545
8	84204	20	33646
10	63643	28	13903
11	109437	31	5999

Theorem 2. *In the source's counting convention,*

$$\sum_{p \le q \in P} n_G(p,q) \;\ge\; 798370 \;>\; 796772.$$

Thus Theorem 1.1(2) of the fixed published source is false.

Proof. The types are invariant under automorphisms, so their orbit sets are disjoint. Sum (1) over the 136 types. The sixteen displayed integer row sums add to 798370. The source asserts the smaller total 796772. The discrepancy is 1598 even with this lower bound; we make no claim that the lower bound is the exact total. □

Theorem 3. *The exact value for type* $(20, 31)$ *is* $n_G(20, 31) = 9118$. *The corresponding entry of the source's Table 1 is 7246.*

Proof. Among the listed maximal subgroups only the two classes of $L_2(31)$ can contain an element of order 31. Their element-order set is $\{1, 2, 3, 4, 5, 8, 15, 16, 31\}$, which excludes 20. This set is read from the retained character tables. It also follows from the usual split, nonsplit and unipotent alternatives in $\mathrm{PSL}_2(31)$, whose nontrivial orders divide 15 or 16, or equal 31. Thus no maximal subgroup contains elements of both prescribed orders, and every pair in $X_G(20, 31)$ generates G. Equation (2) gives

$$x_G(20, 31) = 8403431565957120, \qquad |\operatorname{Aut}(G)| = 921631011840.$$

Their quotient is 9118. The printed entry 7246 was checked directly in the native rendering of source page 11, in addition to extraction. □

Exactly 129 lower bounds in the ledger are positive. Their types are precisely the source's asserted positive types, so their existence is recovered simultaneously. The seven types with zero lower bound are

$$(3, 3), (3, 4), (3, 5), (3, 6), (3, 7), (3, 12), (4, 4).$$

A zero lower bound is not a nonexistence proof. This component does not infer the exclusion of all seven. Twenty of the 136 intervals have equal endpoints, which supplies exact counts for those types; the other intervals are kept as intervals. Apart from $(20, 31)$, every printed entry lies within its calculated interval. That consistency is not a verification of the other 116 unresolved exact counts.

4 Verification, attribution and limits

The executable certificate freshly extracts all character vectors from the named tables and recomputes every rational sum and integer bound. Repeated maximal classes, including the two copies of A_7, are preserved. The source table is used only afterward for comparison. Separate

complete controls in S_3, D_8, A_4, S_4, A_5 enumerate all subgroups and all ordered pairs of each type and check the union inequality and free-orbit divisibility literally.

The proof uses the standard union bound, ordinary character theory and established subgroup classifications. The contribution is their specific application and checked numerical consequence for the fixed source. We claim neither a new maximal-subgroup classification nor priority for any general method. This is a correction to a published assertion, not the settlement of a conjecture. The exact full total, the remaining Table 1 entries and the chirality and inversion spectra remain separate obligations. No Lean verification or numerical compression factor is supplied by this certificate.

References

[1] T. Connor and D. Leemans, *Algorithmic enumeration of regular maps*, Ars Mathematica Contemporanea 10 (2016), 211–222.

[2] R. A. Wilson, *The maximal subgroups of the O'Nan group*, Journal of Algebra 97 (1985), 467–473.

[3] R. A. Wilson, *Maximal subgroups of sporadic groups*, survey dated 8 April 2016, Section 3.3.

[4] R. A. Wilson, S. J. Nickerson, J. N. Bray et al., *Atlas of Group Representations*, O'Nan group page, `https://brauer.maths.qmul.ac.uk/Atlas/v3/spor/ON/`, accessed 13 October 2026.